

Chapter 4: Continuous Distributions

Probability and Statistics for Science and Engineering
with Examples in R

Hongshik Ahn

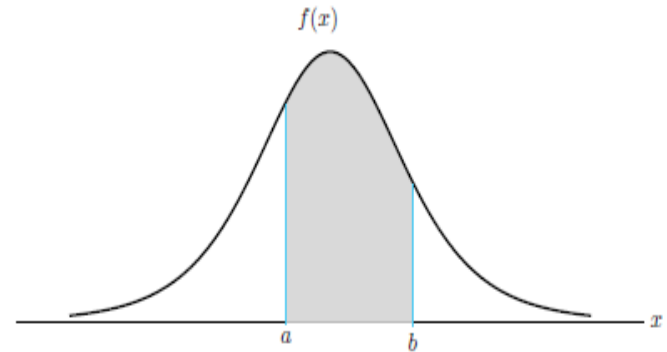
(1) Probability Density

- A r.v. X is continuous if it represents some measurement on a continuous scale.
- Let X be a continuous r.v. Then the probability density function (pdf) $f(x)$ of X has the following properties:

a) $P(a \leq X \leq b) = \int_a^b f(x)dx$ for $a \leq b$

b) $f(x) \geq 0$ for all x

c) $\int_{-\infty}^{\infty} f(x)dx = 1$



- $P(X = c) = \int_c^c f(x)dx = 0$ for any number c
- If X is a continuous r.v., then for any two numbers a & b with $a < b$,
$$P(a \leq X \leq b) = P(a \leq X < b)$$
$$= P(a < X \leq b) = P(a < X < b)$$

(2) Uniform Distribution

Example 4.1: A subway train arrives at a certain stop at 10 minute intervals starting at 6 am. Every time Rachel takes this train, she arrives at the station at a different time. Her waiting time is random within the 10 minute interval.

$$f(x) = \begin{cases} \frac{1}{10}, & 0 \leq x \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

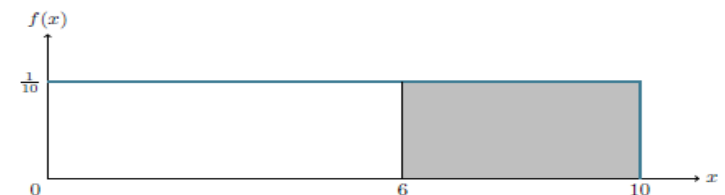
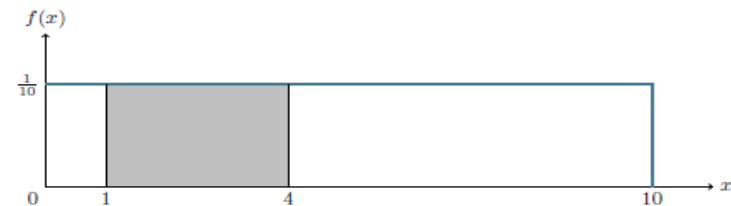
- Probability of waiting between 1 and 4 minutes:

$$P(1 \leq X \leq 4) = \int_1^4 f(x) dx = \int_1^4 \frac{1}{10} dx = \frac{x}{10} \Big|_1^4 = \frac{4}{10} - \frac{1}{10} = \frac{3}{10}$$

$$\text{or } P(1 \leq X \leq 4) = \frac{1}{10} (4 - 1) = \frac{3}{10}$$

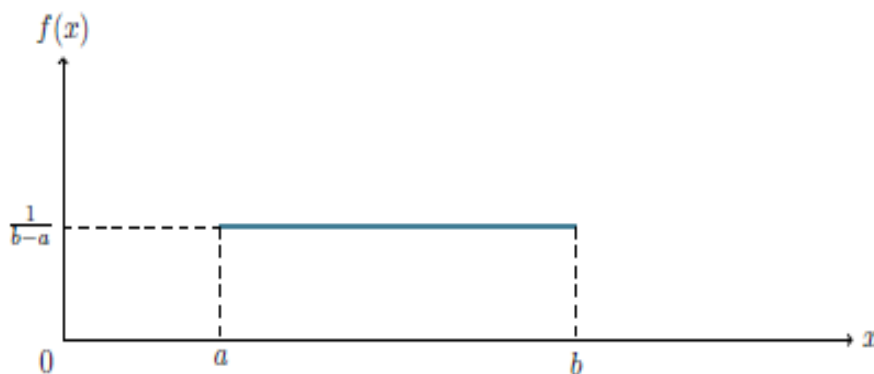
- Probability of waiting for at least 6 minutes:

$$\begin{aligned} P(X \geq 6) &= \int_6^{10} f(x) dx = \int_6^{10} \frac{1}{10} dx = \frac{x}{10} \Big|_6^{10} \\ &= 1 - \frac{6}{10} = \frac{2}{5} \end{aligned}$$



$X \sim \text{Uniform}(a, b)$ if

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$



In the previous example, $a = 0, b = 10$.

Finding Uniform Probabilities Using R

For $\text{Uniform}(a, b)$,

- $P(X < c)$ for $a < c < b$:
 `>punif(c, a, b)`
- $100p^{\text{th}}$ percentile:
 `>qunif(p, a, b)`

For $\text{Uniform}(0, 1)$,

- $P(X < c)$ for $0 < c < 1$:
 `>punif(c)`
- $100p^{\text{th}}$ percentile:
 `>qunif(p)`

For Example 4.1,

$P(1 \leq X \leq 4)$:

`>punif(4, 0, 10)-punif(1, 0, 10)`

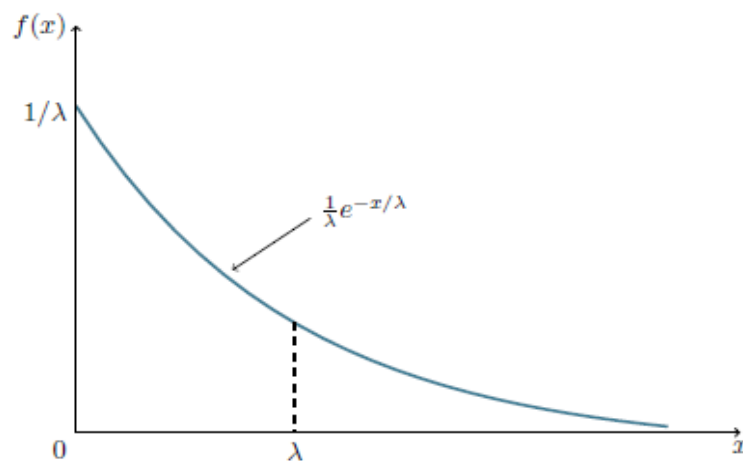
$P(X \geq 6)$:

`>1-punif(6, 0, 10)`

(3) Exponential Distribution

$X \sim \text{Exponential}(\lambda)$ if

- $f(x) = \begin{cases} \frac{1}{\lambda} e^{-x/\lambda}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$ where $\lambda > 0$



$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} \frac{1}{\lambda} e^{-x/\lambda} dx = -e^{-x/\lambda} \Big|_0^{\infty} = -(0 - 1) = 1$$

Example 4.2: The lifetime (in years) of a certain rodent species:

$$f(x) = \begin{cases} \frac{1}{3} e^{-x/3}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

a) Probability that a rodent in this species lives at least 2 years?

$$P(X > 2) = \int_2^{\infty} f(x) dx = \int_2^{\infty} \frac{1}{3} e^{-x/3} dx = -e^{-x/3} \Big|_2^{\infty} = e^{-2/3} = 0.5134$$

b) Probability that a rodent in this species lives between 1 and 2 years?

$$P(1 < X < 2) = \int_1^2 \frac{1}{3} e^{-x/3} dx = -e^{-x/3} \Big|_1^2 = e^{-1/3} - e^{-2/3} = 0.2031$$

Finding Exponential Probabilities Using R

For $\text{Exponential}(\lambda)$,

- $P(X < c) \ c > 0$:
 `>pexp(c, 1/λ)`
- $100p^{\text{th}}$ percentile:
 `>qexp(p, 1/λ)`

For Example 4.2,

$P(X \geq 2)$:

`>1-pexp(2, 1/3)`

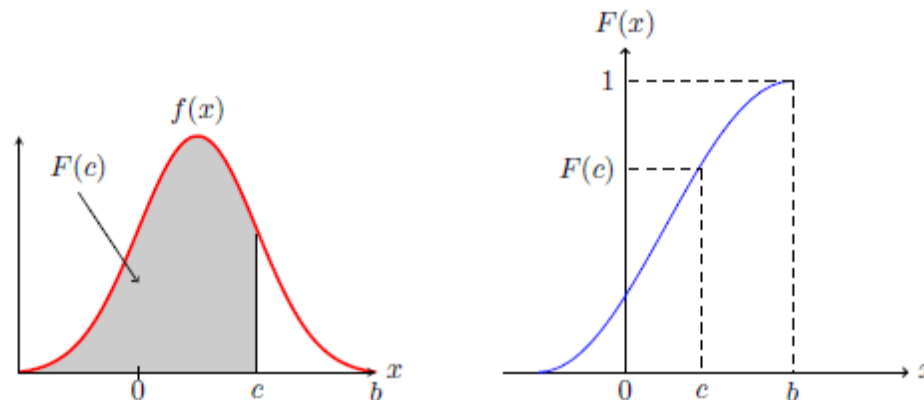
$P(1 \leq X \leq 2)$:

`>pexp(2,1/3)-pexp(1, 1/3)`

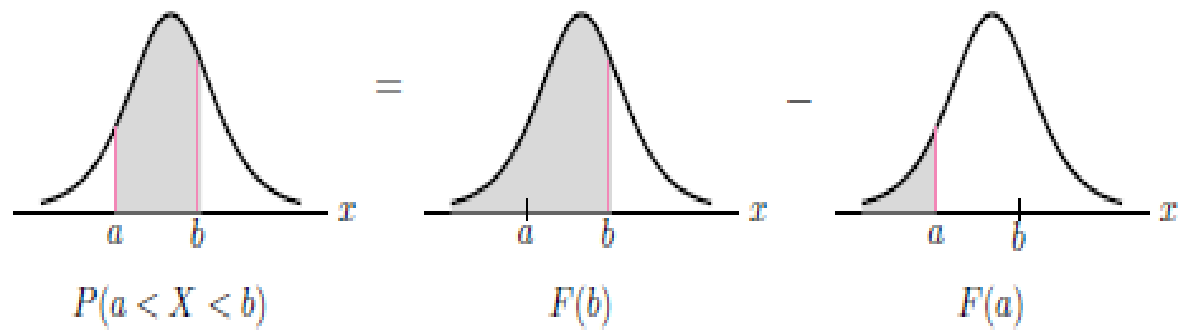
(4) Cumulative Distribution function

- CDF (Cumulative distribution function) $F(x)$ for a continuous r.v. X :

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$$



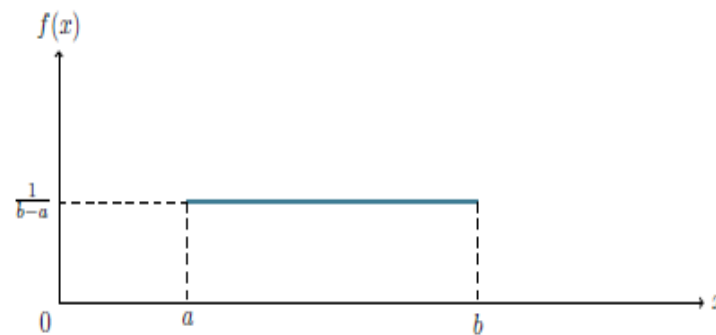
- $P(a \leq X \leq b) = F(b) - F(a)$



- $P(X \geq a) = 1 - P(X \leq a) = 1 - F(a)$

Example 4.3: $X \sim \text{Uniform}(a, b)$. Find the cdf.

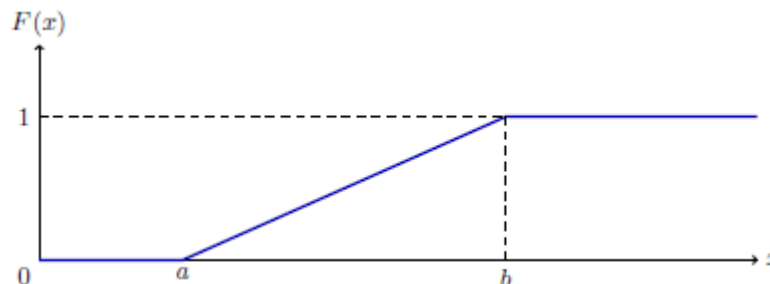
$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$



For $a \leq x \leq b$,

$$F(x) = \int_{-\infty}^x f(y) dy = \int_a^x \frac{1}{b-a} dy = \frac{y}{b-a} \Big|_a^x = \frac{x-a}{b-a}$$

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}$$



Example 4.4: A pdf of X is given as

$$f(x) = \begin{cases} \frac{1}{\lambda} e^{-x/\lambda}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Then the cdf is

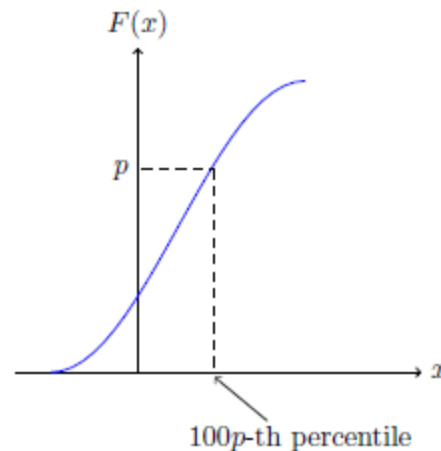
$$F(x) = \begin{cases} \int_{-\infty}^x f(y) dy = \int_0^x \frac{1}{\lambda} e^{-y/\lambda} dy = -e^{-\frac{y}{\lambda}} \Big|_0^x = 1 - e^{-\frac{x}{\lambda}}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

If X is a continuous r.v., then

$F'(x) = f(x)$ at every x at which $F'(x)$ exists.

Percentile

- The $100p$ -th percentile: x such that $F(x) = p$
- Median: x such that $F(x) = \frac{1}{2}$



Example 4.5: Find the 75th percentile of the following distribution.

$$f(x) = \begin{cases} 3x^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

For $0 \leq x \leq 1$,

$$F(x) = \int_{-\infty}^x f(y) dy = \int_0^x 3y^2 dy = y^3 \Big|_0^x = x^3$$

The 75th percentile:

$$F(x) = \frac{3}{4} \rightarrow x^3 = \frac{3}{4} \rightarrow x = \sqrt[3]{3/4}$$

(5) Expectations

- Expected value (mean) of a continuous r.v.:

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

- Variance of a continuous r.v.:

$$\sigma^2 = Var(X) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx$$

- Standard Deviation of X: $\sigma = \sqrt{Var(X)}$
- $Var(X) = E(X^2) - [E(X)]^2$

Example 4.6:

The distribution of the amount of gravel (in tons) sold by a particular construction supply company in a given week is continuous with pdf

$$f(x) = \begin{cases} 6x(1-x), & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx = \int_0^1 6x^2(1-x)dx = 6 \int_0^1 (x^2 - x^3)dx \\ &= 6 \left(\frac{x^3}{3} - \frac{x^4}{4} \right) \Big|_0^1 = 6 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^1 6x^3(1-x)dx \\ &= 6 \int_0^1 (x^3 - x^4)dx = 6 \left(\frac{x^4}{4} - \frac{x^5}{5} \right) \Big|_0^1 = 6 \left(\frac{1}{4} - \frac{1}{5} \right) = \frac{3}{10} \end{aligned}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{3}{10} - \left(\frac{1}{2} \right)^2 = \frac{1}{20}$$

Example 4.7: $X \sim \text{Uniform}(a, b)$

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx = \int_a^b \frac{x}{b-a} dx = \frac{x^2}{2(b-a)} \Big|_a^b = \frac{b^2 - a^2}{2(b-a)} \\ &= \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x)dx = \int_a^b \frac{x^2}{b-a} dx = \frac{x^3}{3(b-a)} \Big|_a^b = \frac{b^3 - a^3}{3(b-a)} \\ &= \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} = \frac{a^2 + ab + b^2}{3} \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{4(a^2 + ab + b^2) - 3(a+b)^2}{12} = \frac{a^2 - 2ab + b^2}{12} = \frac{(a-b)^2}{12} \end{aligned}$$

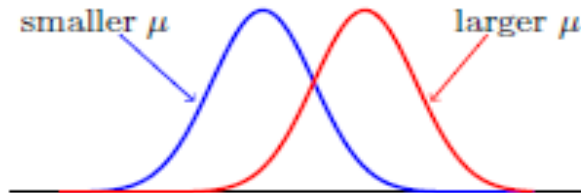
(6) Normal Distribution

$X \sim N(\mu, \sigma^2)$:

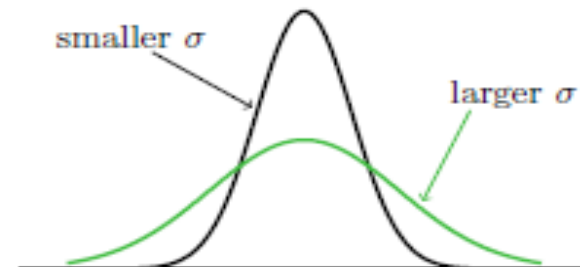
$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

$$E(X) = \mu, \quad \text{Var}(X) = \sigma^2$$

Same sd (σ), different mean (μ)



Same μ , different σ



Standard Normal Distribution

$$Z \sim N(0,1) \quad \leftarrow \quad \mu = 0, \sigma^2 = 1$$
$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

$$P(Z \leq z) = \int_{-\infty}^z f(x) dx = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \Phi(z)$$

$$\Phi(z) = 1 - \Phi(-z)$$

$$P(Z \geq z) = P(Z \leq -z)$$

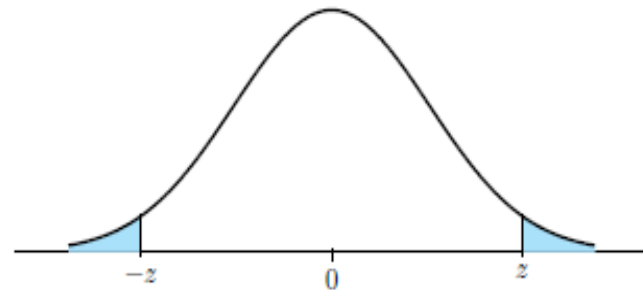
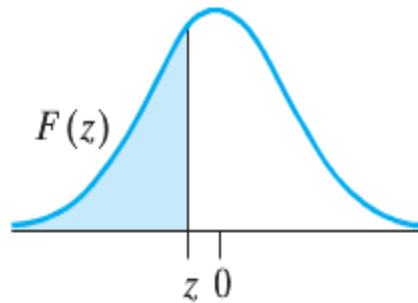


Table A.1 Standard Normal Distribution Table

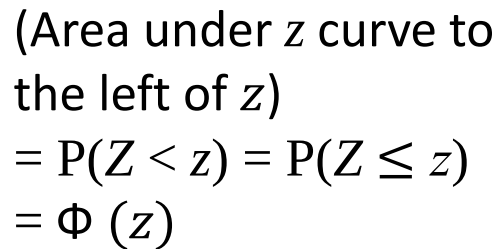


(Area under z curve
to the left of z)
 $= P(Z < z) = P(Z \leq z)$
 $= \Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
z	Area									
-3.50	0.00023263									
-4.00	0.00003167									
-4.50	0.00000340									
-5.00	0.00000029									

Source: Computed by M. Longnecker using Splus.

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3	0.9995	0.9995	0.99							



Two Applications

- **Find the probability:**

Using the standard normal table, find the following probability for z_0 from a standard normal distribution:

$$P(Z \leq z_0) = ?$$

- **Find the percentile:**

Using the standard normal table, find the z value that satisfies

$$P(Z \leq ?) = p_0$$

Finding Normal Probabilities Using R

For $N(\mu, \sigma^2)$,

- cdf:
 `>pnorm(x,  $\mu$ ,  $\sigma$ )`
- 100 p -th percentile:
 `>qnorm(p,  $\mu$ ,  $\sigma$ )`

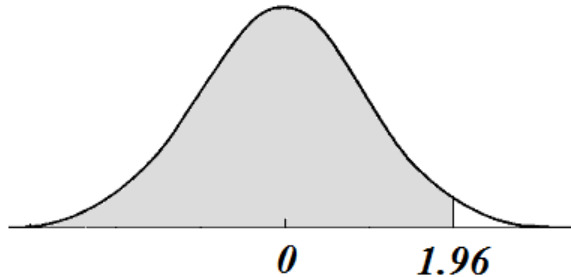
For $N(0,1)$,

- cdf:
 `>pnorm(x)`
- 100 p -th percentile:
 `>qnorm(p)`

Example 4.8: Finding probabilities

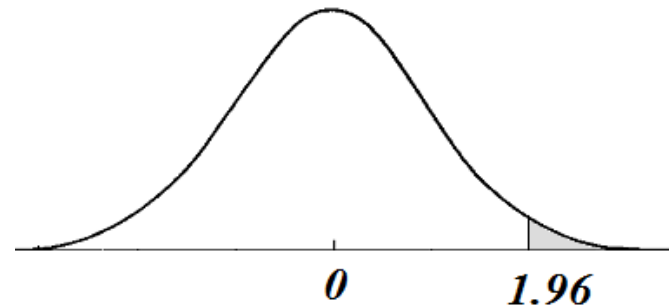
$$P(Z < 1.96) = 0.9750$$

>pnorm(1.96)



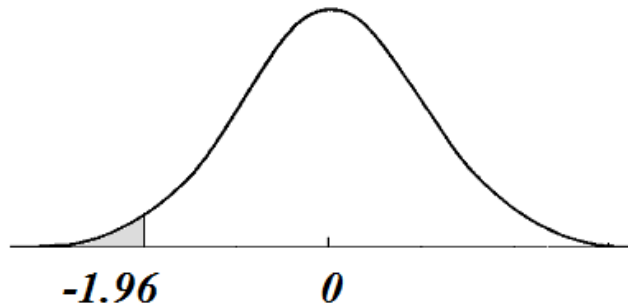
$$P(Z > 1.96) = 1 - \Phi(1.96)$$
$$= 1 - 0.9750 = 0.025$$

>1-pnorm(1.96)



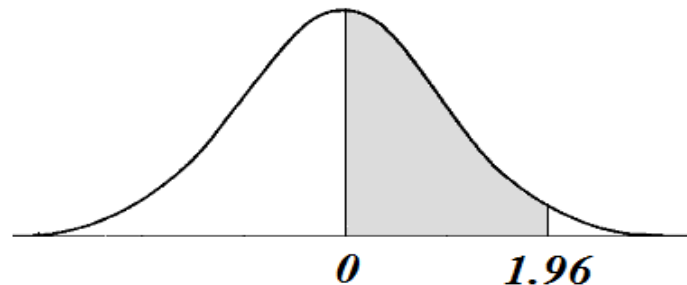
$$P(Z \leq -1.96) = 0.025$$

>pnorm(-1.96)



$$P(0 \leq Z \leq 1.96)$$
$$= \Phi(1.96) - \Phi(0)$$
$$= 0.9750 - 0.5 = 0.475$$

>pnorm(1.96)-pnorm(0)



Percentiles

Example 4.9

- $P(Z \leq ?) = p$

$$P(Z \leq z) = 0.937 \Rightarrow z = 1.53$$

`>qnorm(.937)`

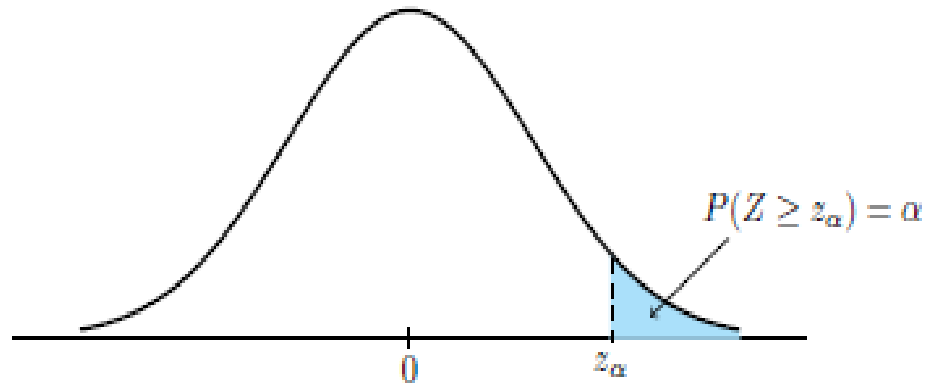
$$\begin{aligned} P(Z \geq z) = 0.4920 &\Rightarrow 1 - P(Z \leq z) = 0.492 \\ &\Rightarrow P(Z \leq z) = 0.508 \Rightarrow z = 0.02 \end{aligned}$$

`>qnorm(0.508)`

Z_α

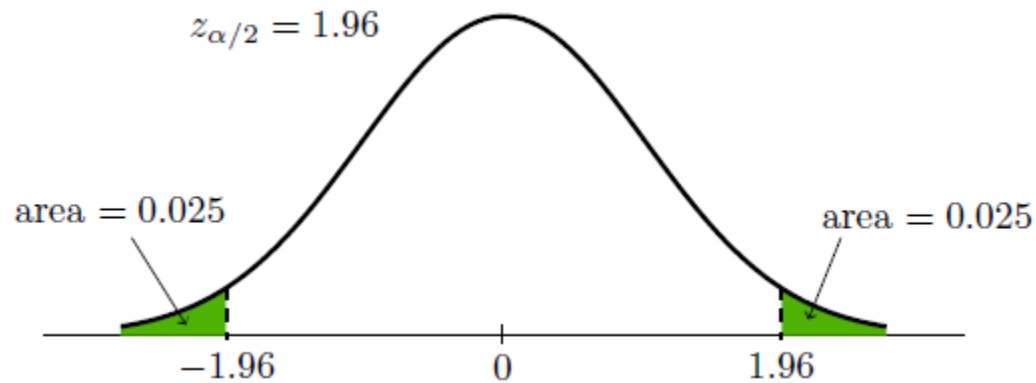
The probability that z_α will be exceeded by a random variable having the standard normal distribution is α .

That is, $\alpha = P(Z > z_\alpha)$, or $z_\alpha = 100(1 - \alpha)$ th Percentile.



Percentile	90	95	97.5	99	99.5
α (right tail area)	.1	.05	.025	.01	.005
z_α	1.28	1.645	1.96	2.33	2.58

If $z_{\alpha/2} = 1.96$, the shaded area is 0.05.



$$X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$P(a \leq X \leq b) = P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$$

Example 4.10: $X \sim N(50, 25)$

$$\begin{aligned} P(44 \leq X \leq 53) &= P\left(\frac{53 - 50}{5} \leq Z \leq \frac{44 - 50}{5}\right) \\ &= P(-1.2 \leq Z \leq 0.6) = \Phi(0.6) - \Phi(-1.2) \\ &= 0.7257 - 0.1151 = 0.6106 \end{aligned}$$

Example 4.11: The raw scores on a SAT math test are normally distributed with mean 510 and sd 110.

(a) What proportion of the candidates score below 620?

$$P(X < 620) = P\left(Z < \frac{620 - 510}{110}\right) = \Phi(1) = 0.8413$$

`>pnorm(620, 510, 110)`

(b) Find the 42nd percentile of the scores.

$$\Phi(z) = 0.42 \rightarrow z = -0.20$$

$$\begin{aligned} z &= \frac{x - 510}{110} \Rightarrow x = 110z + 510 \\ &= 110(-0.2) + 510 = 488 \end{aligned}$$

`>qnorm(0.42, 510, 110)`

Example 4.12:

$$P(X \text{ is within 1 sd of its mean}) = P(\mu - \sigma \leq X \leq \mu + \sigma)$$

$$= P\left(\frac{\mu - \sigma - \mu}{\sigma} \leq Z \leq \frac{\mu + \sigma - \mu}{\sigma}\right) = P(-1 \leq Z \leq 1)$$

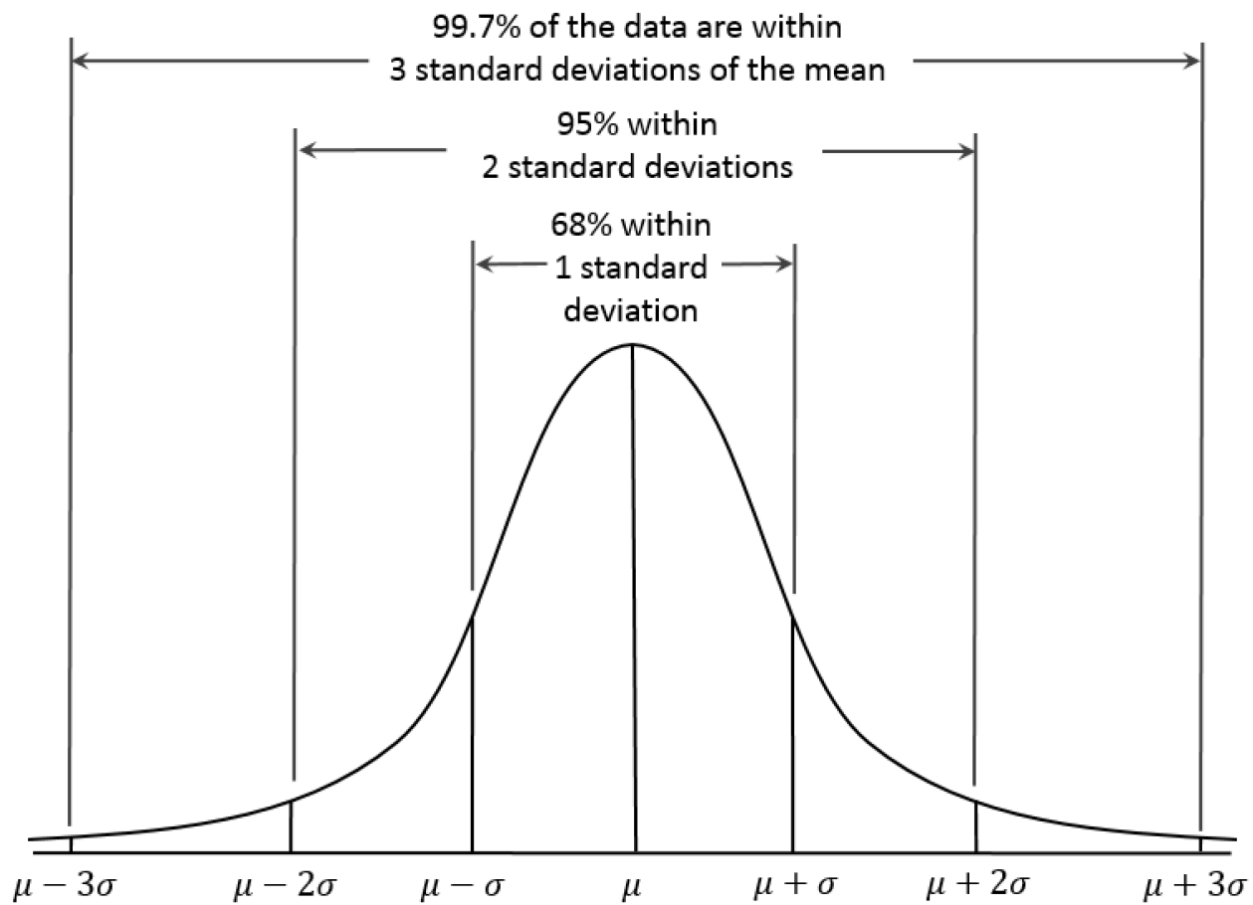
$$= \Phi(1) - \Phi(-1) = 0.8413 - 0.1587 = 0.6826$$

$$P(X \text{ is within 2 sd of its mean}) = P(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$$

$$= \Phi(2) - \Phi(-2) = 0.9772 - 0.0228 = 0.9544$$

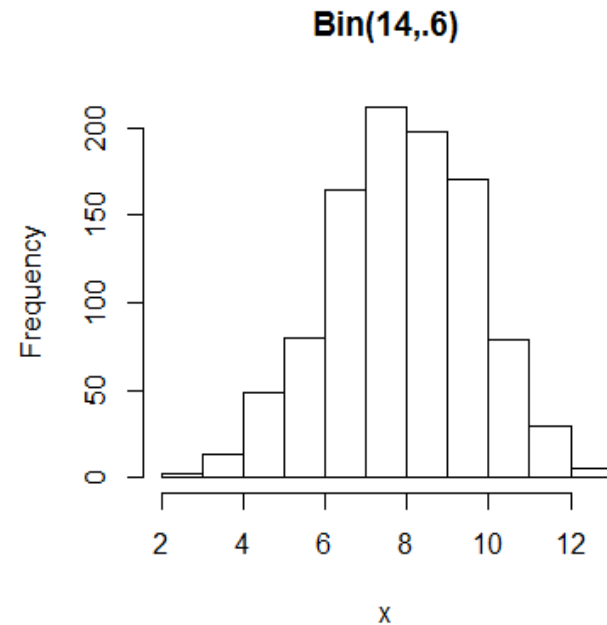
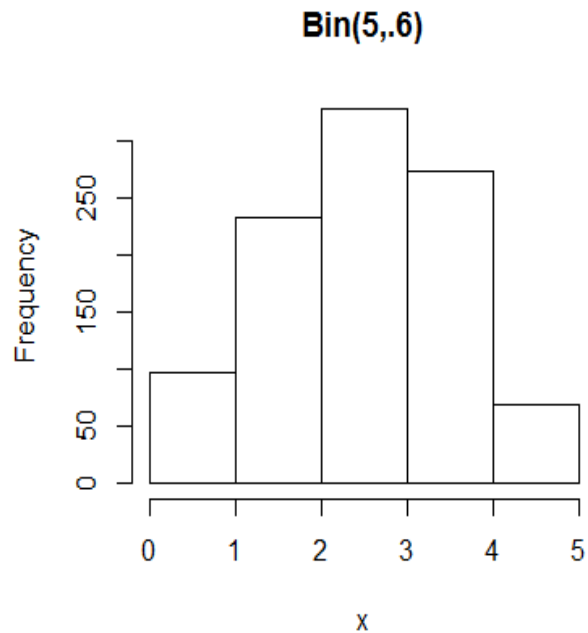
$$P(X \text{ is within 3 sd of its mean}) = P(\mu - 3\sigma \leq X \leq \mu + 3\sigma)$$

$$= \Phi(3) - \Phi(-3) = 0.9987 - 0.0013 = 0.9974$$



Normal Approximation to the Binomial

$$X \sim \text{Bin}(n, p)$$



Let $X \sim \text{Bin}(n, p)$.

When np and $n(1 - p)$ are both large (≥ 10),

$Z = \frac{X - np}{\sqrt{np(1-p)}}$ is approximately $N(0,1)$.

The method of approximation:

Step 1 Find $\mu = np$ and $\sigma^2 = np(1 - p)$

Step 2 Use $N(\mu, \sigma^2) = N(np, np(1 - p))$.

Step 3 Remember to use continuity correction.

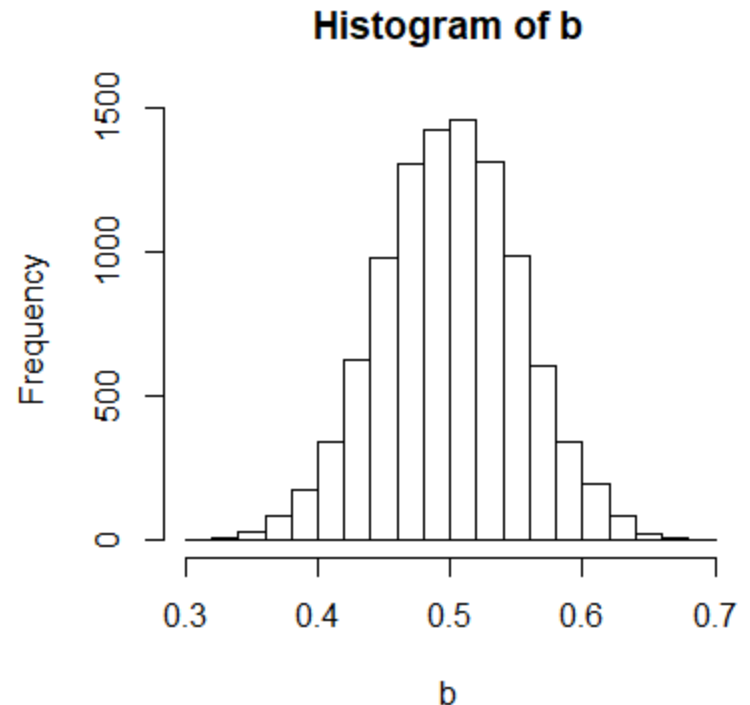
Example: Generate 100 Unif(-10,10) random numbers

$$u_{1,1}, u_{1,2}, \dots, u_{1,100}.$$

Take the mean

$$x_1 = \bar{u}_1 = \frac{\sum_{i=1}^{100} u_{1,i}}{100}.$$

Histogram of $(x_1, x_2, \dots, x_{100})$



Example 4.13: Practice of continuity correction

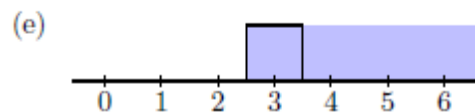
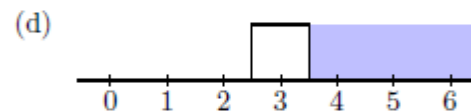
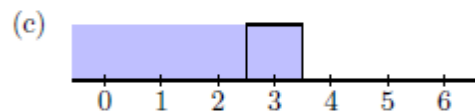
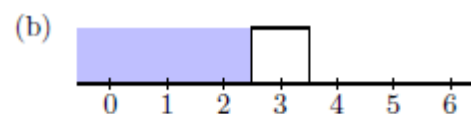
$$(a) \quad P(X = 3) = P\left(\frac{2.5 - \mu}{\sigma} < Z < \frac{3.5 - \mu}{\sigma}\right)$$

$$(b) \quad P(X < 3) = P\left(Z < \frac{2.5 - \mu}{\sigma}\right)$$

$$(c) \quad P(X \leq 3) = P\left(Z < \frac{3.5 - \mu}{\sigma}\right)$$

$$(d) \quad P(X > 3) = P\left(Z > \frac{3.5 - \mu}{\sigma}\right)$$

$$(e) \quad P(X \geq 3) = P\left(Z > \frac{2.5 - \mu}{\sigma}\right)$$



Example 4.14: Let $X \sim \text{Bin}(n, p)$, $n = 30$, $p = 0.4$.

Find $P(X = 14)$ in the following ways:

(a) using the exact distribution

$$P(X = 14) = \binom{30}{14} (0.4)^{14} (0.6)^{16} = 0.1101$$

(b) Using a normal approximation

$$\mu = np = 30(0.4) = 12, n(1 - p) = 30(0.6) = 18$$

$$\sigma = \sqrt{np(1 - p)} = \sqrt{30 \cdot 0.4 \cdot 0.6} = \sqrt{7.2} = 2.6833$$

$$\begin{aligned} P(13.5 \leq X \leq 14.5) &= P\left(\frac{13.5 - 12}{2.6833} \leq Z \leq \frac{14.5 - 12}{2.6833}\right) \\ &= P(0.56 \leq Z \leq 0.93) \approx \Phi(0.93) - \Phi(0.56) \\ &= 0.8238 - 0.7123 = 0.1115 \end{aligned}$$

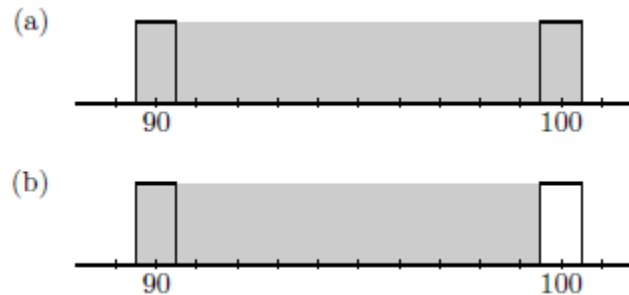
Example 4.15: Let $X \sim \text{Bin}(n, p)$, $n = 160$, $p = 0.6$.

$$\mu = np = 160(0.6) = 96, n(1 - p) = 160(0.4) = 64,$$

$$\sqrt{160(0.6)(0.4)} = \sqrt{38.4} = 6.197$$

$$\begin{aligned} \text{(a)} \quad P(90 \leq X \leq 100) &= P\left(\frac{89.5-96}{6.197} \leq Z \leq \frac{100.5-96}{6.197}\right) \\ &= P(-1.05 \leq Z \leq 0.73) \approx \Phi(0.73) - \Phi(-1.05) \\ &= 0.7673 - 0.1469 = 0.6204 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(90 \leq X < 100) &= P\left(\frac{89.5-96}{6.197} \leq Z \leq \frac{99.5-96}{6.197}\right) \\ &= P(-1.05 \leq Z \leq 0.56) \approx \Phi(0.56) - \Phi(-1.05) \\ &= 0.7123 - 0.1469 = 0.5654 \end{aligned}$$



Example 4.16: In a certain country, 30% of the male adult population smoke regularly. In a random sample of 750 adults, what is the probability that (a) less than 200 and (b) 240 or more are smokers?

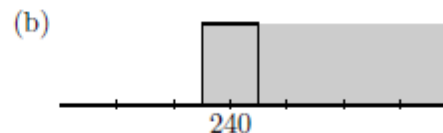
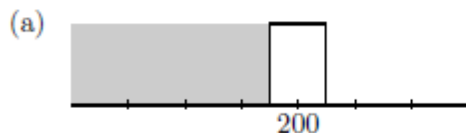
Let X be the number of smokers in the random sample.
Then $X \sim \text{Bin}(750, 0.3)$.

$$\mu = np = 750(0.3) = 225,$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{750(0.3)(0.7)} = \sqrt{157.5} = 12.55$$

$$(a) \quad P(X < 200) = P\left(Z \leq \frac{199.5 - 225}{12.55}\right) = \Phi(-2.03) = 0.0212$$

$$(b) \quad P(X \geq 240) = P\left(Z \geq \frac{239.5 - 225}{12.55}\right) = P(Z \geq 1.16) \\ \approx 1 - \Phi(1.16) = 1 - 0.8770 = 0.1230$$



(7) Gamma Distribution

For $\alpha > 0$, the gamma function $\Gamma(\alpha)$ is defined by

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx.$$

1. For any $\alpha > 1$, $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$
2. For any positive integer n , $\Gamma(n) = (n - 1)!$
3. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

If $X \sim \text{Gamma}(\alpha, \beta)$,

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, & x \geq 0, \alpha > 0, \beta > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2$$

Finding Gamma Probabilities Using R

For $\text{Gamma}(\alpha, \beta)$,

- $P(X < c) \ c > 0$:
 `>pgamma(c,  $\alpha$ ,  $1/\beta$ )`
- 100 p -th percentile:
 `>qgamma(p,  $\alpha$ ,  $1/\beta$ )`

Example 4.17: Let X be a random variable representing the load of a web server. Suppose X is distributed as gamma with $\alpha = 1$ and $\beta = 5$.

a) Find $P(X < 2)$.

$$f(x) = \begin{cases} \frac{1}{\Gamma(1)5^1} x^{1-1} e^{-x/5} = \frac{1}{5} e^{-x/5} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X < 2) = \int_0^2 \frac{1}{5} e^{-x/5} dx = -e^{-x/5} \Big|_0^2 = 1 - e^{-2/5} = 1 - 0.6703 = 0.3297$$

b) Find $P(1 < X < 3)$.

$$P(1 < X < 3) = \int_1^3 \frac{1}{5} e^{-x/5} dx = -e^{-x/5} \Big|_1^3 = e^{-1/5} - e^{-3/5} = 0.2699$$

c) Find the mean and variance of X .

$$E(X) = \alpha\beta = 1(5) = 5$$

$$Var(X) = \alpha\beta^2 = 1(5^2) = 25$$

If $\alpha = 1$,

$$X \sim \text{Exponential}(\beta) = \text{Gamma}(1, \beta)$$

$$f(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0, \beta > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \beta, \quad \text{Var}(X) = \beta^2$$

$$F(x) = \begin{cases} 1 - e^{-x/\beta}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- χ^2 distribution with degrees of freedom (df) ν $\left(\text{Gamma}\left(\frac{\nu}{2}, 2\right)\right)$:

$$f(x) = \begin{cases} \frac{1}{\Gamma\left(\frac{\nu}{2}\right) 2^{\nu/2}} x^{\frac{\nu}{2}-1} e^{-x/2}, & x > 0, \nu > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \alpha\beta = \frac{\nu}{2} \cdot 2 = \nu$$

$$\text{Var}(X) = \alpha\beta^2 = \frac{\nu}{2} \cdot 4 = 2\nu$$

(8) Beta Distribution

If $X \sim \text{Beta}(\alpha, \beta)$,

$$f(x) = \begin{cases} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, & 0 < x < 1, \alpha > 0, \beta > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

In R,

- $P(X \leq c)$:
 `>pbeta(c,  $\alpha$ ,  $\beta$ )`
- 100 p -th percentile:
 `>qbeta(p,  $\alpha$ ,  $\beta$ )`

Example 4.17: At a gas station, the fraction of customers who purchase supreme gas in a week is a r.v. with Beta(1, 4).

- (a) If the gas station serves an average of 850 customers per week, how many customers are expected to purchase supreme gas next week?

$$E(X) = \frac{\alpha}{\alpha + \beta} = \frac{1}{1 + 4} = 0.2$$

The expected number of such customers is $850(0.2) = 170$.

- (b) What is the probability that more than 25% of the customers will purchase supreme gas over the next week?

$$f(x) = \frac{\Gamma(1+4)}{\Gamma(1)\Gamma(4)} x^{1-1} (1-x)^{4-1} = \frac{4!}{3!} (1-x)^3 = 4(1-x)^3, 0 < x < 1$$

$$P(X > 1/4) = \int_{1/4}^1 4(1-x)^3 dx = -(1-x)^4 \Big|_{1/4}^1 = \left(\frac{3}{4}\right)^4 = \frac{81}{256} = 0.3164$$

Using R,

`>1-pbeta(1/4, 1, 4)`